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Total Number of Pages : 02

B.Tech
PET3I103

3rd Semester Regular / Back Examination 2018-19

SIGNAL & SYSTEMS

BRANCH : ECE, ETC

Time : 3 Hours

Max Marks : 100

Q.CODE : E938

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Short Answer Type Questions (Answer All-10) (2 x 10)

- List the ROC properties of Laplace transform.
- Find the Z-transform of a sequence $x[n] = \cos(n\omega T) u[n]$.
- Find the Fourier transform of $x(t) = e^{-at}u(t)$
- Is there be two different signals having same Laplace transform? Give an example. How do you differentiate these two signals?
- Define the convolutional integral.
- What is aliasing?
- Given $x(n) = \{1, -4, 3, 1, 5, 2\}$. Represent $x(n)$ in terms of weighted shifted impulse function.
- State the multiplication property of DTFT.
- Using Z-transform check whether the following system is stable or not?

$$H(z) = \frac{z}{z - 1/2} + \frac{2z}{z + 3}, \quad \frac{1}{2} < |z| < 3$$

- State and prove the time folding property of Z- transform.

Part- II

Q2 Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Check the system $y(n) = \log_{10}(x[n])$ is linear, time invariant, causal and static.
- Find out whether the following signal is energy or power signal or neither any of two. Determine power or energy if found.
 $x(t) = u(t) + 5u(t - 1) - 2u(t - 2)$
- Find the spectrum of $x(t) = e^{-2|t|}$. Plot the spectrum of the signal.
- Find the Fourier transform of a rectangular pulse with width T and amplitude A.
- State and prove Parseval's theorem of Fourier Transform.
- Find the inverse Laplace transform of $X(s) = \frac{1}{(s+5)(s-3)}$ for the ROC $-5 < \text{Re}\{s\} < 3$
- Using Laplace transform of $x(t)$. Give the pole-zero plot and find ROC of the signal $x(t)$.
 $x(t) = e^{-b|t|}$ for both $b > 0$ and $b < 0$.
- Find the Z-transform and sketch the ROC of the following sequence
 $x[n] = 2^n u[n] + 3^n u(-n - 1)$

- Using the properties of inverse Z-transform solve

$$X(z) = \log(1 + az^{-1}); |z| > |a| \text{ and } X(z) = \frac{az^{-1}}{(1 - az^{-1})^2}; |z| > |a|$$

- Determine DTFT of $\left(\frac{1}{2}\right)^n u(n)$. Plot its spectrum.

- Using convolution property of DTFT, find the inverse DTFT of

$$X(e^{j\omega}) = \frac{1}{(1 - ae^{j\omega})^2}, |a| < 1$$

- Find the convolution sum of $x[n] = r[n]$ and $h[n] = u[n]$.

Part-III

Long Answer Type Questions (Answer Any Two out of Four)

- Q3

Using convolution integral, determine the response of a CTLTI system $y(t)$ given input $x(t) = e^{-\alpha t}u(t)$ and impulse response $h(t) = e^{-\beta t}u(t)$, $|\alpha| < 1, |\beta| < 1$

(16)
- Q4

Analyze on recursive and non-recursive systems with an example

(16)
- Q5

a) Using graphical method, find the output sequence $y[n]$ of the LTI system whose response $h[n]$ is given and input $x[n]$ is given as follows. $x[n] = \{0,5,2\}$ and $h[n] = \{1,1,1\}$.

b) Consider an analog signal $x(t) = 5 \cos 200\pi t$.

i. Determine the minimum sampling rate to avoid aliasing.

ii. If sampling rate is 400Hz, What is the DT signal after sampling?

(16)
- Q6

a) State and prove Sapling Theorem.

b) State and prove following properties of DTFT.

i. Differentiation in frequency

ii. Convolution in frequency domain

(16)